

BIOMECHANICS (EPHE 341)

Angular kinematics

McGinnis Chpt 6, pg 147–158

By the end of this lecture you should be able to...

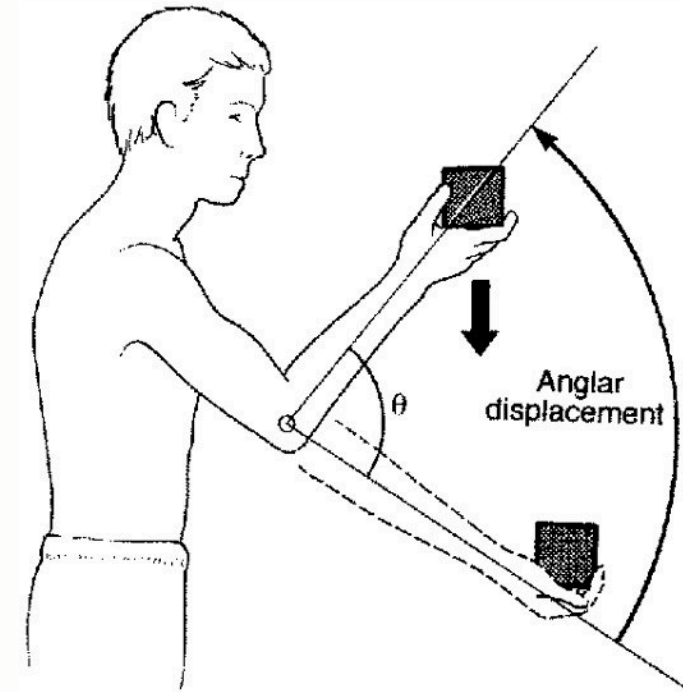
- **Distinguish** absolute from relative angles, and distance from displacement
- **Define** angular displacement, velocity and acceleration, with their units
- **Calculate** radial and total linear acceleration
- **Convert** between degrees, radians and revolutions
- **Relate** angular to linear quantities through the radius:
 $l = r\theta$, $v_t = r\omega$, $a_t = \alpha r$
- **Explain** how a gyroscope measures angular velocity, and convert its output to a speed

What angular kinematics is

Motion about an axis, without asking what caused it

- The study of motion of an object **about a circular path**, exclusive of the influences of mass and force
- It includes **angular displacement**, **angular velocity** and **angular acceleration**
- Angular motion occurs when **all points on an object move in a circular path around the same axis**

"Kinematics" means we describe the motion and say nothing about what produced it. Forces and moments arrive in Angular Kinetics; this lecture is geometry and clocks.



Part 01

Angular displacement

Angles, radians, and the link to distance travelled

The angle moved through, from a starting position

$$\theta = \text{angle}_2 - \text{angle}_1$$

- Defined relative to an **axis of rotation**
- A **vector**: it has magnitude and direction
- **Theta (θ)** represents angular displacement
- SI unit: the **radian**

Measured against what?

ABSOLUTE ANGLES

- Measured from an **external frame of reference** — one that does not move
- Usually from the vertical or the horizontal
- Describes the **orientation of a segment in space**

RELATIVE ANGLES

- The angle formed **between two limb segments**
- Measured against a line that is itself capable of moving
- By convention, in the anatomical position all relative joint angles are **0°**

A joint angle is relative. A segment angle is absolute. Reporting one when you mean the other is the most common error in a gait report.

Absolute and relative angles

Move the thigh and watch which numbers change



thigh 15° from vertical · shank -30° from vertical · knee angle 135° · turn the whole leg and the two absolute angles both change while the knee angle does not — that is the difference

both

absolute only

relative only

Angular distance and angular displacement

The same scalar-and-vector pairing as in linear kinematics

	ANGULAR DISTANCE Φ	ANGULAR DISPLACEMENT Θ
What it is	Length of the angular pathway, start to finish	Angle between the initial and final positions
Type	Scalar — magnitude only	Vector — magnitude and direction
SI unit	radians	radians
Equation	no simple equation	$\theta = \text{angle}_2 - \text{angle}_1$

A gymnast who completes a full backward giant has turned through an angular **distance** of 2π but an angular **displacement** of zero — they finish where they started.

Units of angular measurement

Three in common use, and only one of them works in equations

Revolution	Angular motion that proceeds in one direction and returns to the start
Degree	One revolution contains 360 degrees
Radian	The ratio of the length of an arc to the length of the radii that enclose it

Every equation in this lecture needs radians. Degrees are for reporting; revolutions are for counting; radians are for calculating.

What is a radian?

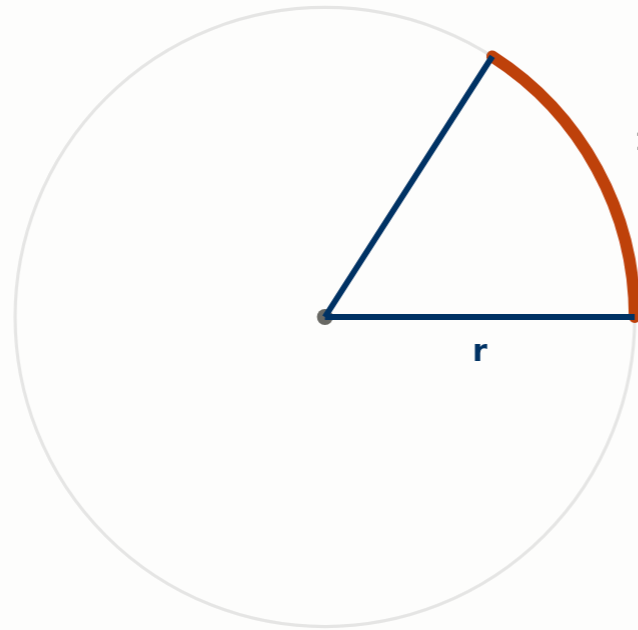
The angle subtended by an arc one radius long

$$\theta = l / r$$

- **1 rad**: the angle at the centre of a circle subtended by an arc equal in length to the radius
- A whole revolution is the circumference over the radius, $2\pi r / r = 2\pi$
- $1 \text{ rad} \approx 57.3^\circ$
- Because it is a ratio of two lengths, a radian has **no units**

What is a radian?

Lay the radius along the rim and count how many fit



$$\theta = l / r$$

$$1.00 \text{ rad} = 57.3^\circ$$

$$\text{a whole turn is } 2\pi r / r = 2\pi \approx 6.28 \text{ rad}$$

the arc, straightened



one radius

$\theta = 1.00 \text{ rad} = 57.3^\circ$ · arc length $1.00 \times r$ · a radian is a ratio of two lengths, so it has no units at all — which is why $r\theta$ comes out in metres

1 rad

$\pi/2$

π

2π

Angle



1.00 rad

Converting degrees to radians

One relationship, used in both directions

$$? \text{ radians} = \text{degrees} \times \frac{\pi \text{ radians}}{180 \text{ degrees}}$$

Multiply the number of degrees by π and divide by 180.

Example: what is 45° in radians?

$$\begin{aligned} &45 \times \pi / 180 \\ &= \mathbf{0.785 \text{ radians}} \end{aligned}$$

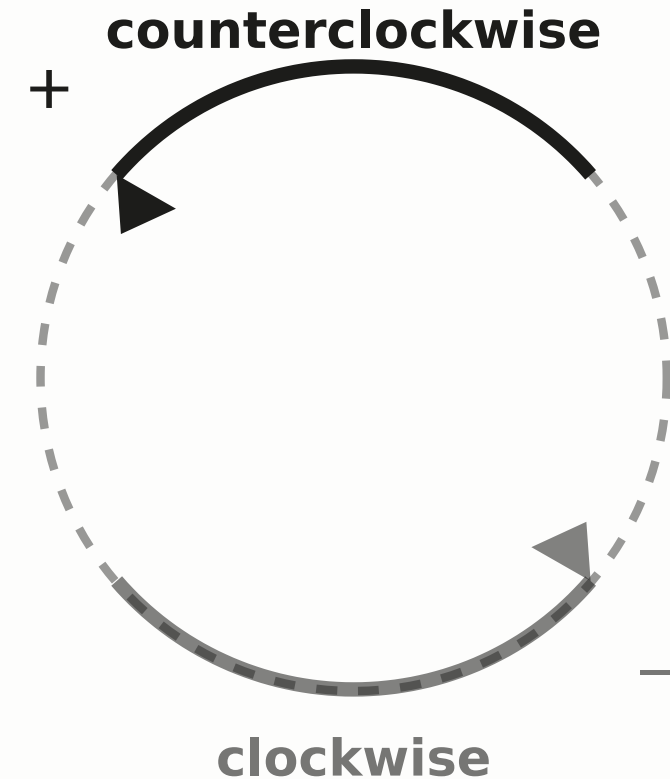
1 rad	$360^\circ / 2\pi = 57.3^\circ$
$\pi/2$ rad	90°
π rad	180°
2π rad	360°
1°	$2\pi / 360 \text{ rad}$

Direction conventions

Angular quantities are vectors, so they need a sign

- Angular displacement, velocity and acceleration are all **vector quantities** — magnitude and direction
- The direction is either **positive** or **negative**, set by the **right hand rule**
- **Counterclockwise is positive**
- **Clockwise is negative**

Curl the fingers of your right hand the way the object is turning; your thumb points along the vector. For motion in the plane of the page that thumb points out of the page for positive, into it for negative.



Angular and linear displacement

The angle is shared; the distance is not

- The angular displacement of **all points** along a rotating object is the **same θ** , however large the radius
- But the **linear path** each point takes depends on how far it is from the axis
- A point twice as far out travels twice as far

$$l = r\theta$$

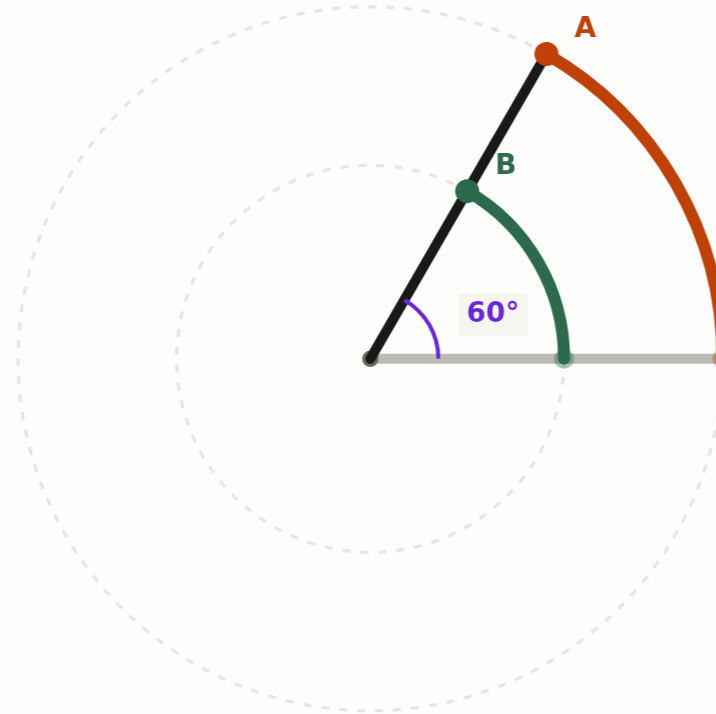
l is the curvilinear distance (m), r the radius of rotation (m), θ the angular displacement **in radians**



The hands and the bat tip share θ — not l

Angular and linear displacement

One angle, two points, two distances



$$l = r\theta$$

point A, $r = 1.00 \text{ m}$

$l = 1.05 \text{ m}$

point B, $r = 0.55 \text{ m}$

$l = 0.58 \text{ m}$

both points turn through **$60^\circ = 1.05 \text{ rad}$** · A travels **$1.05 \text{ m}$** , B travels **$0.58 \text{ m}$** · same θ , different l — the radius is the only thing that differs, and θ must be in radians for $r\theta$ to work

Angle turned



60°

Example Problem

What is the linear displacement of a point on a bicycle wheel (radius = 34 cm) if the wheel rotates 30° ?

Given:

$$r = 0.34 \text{ m}$$

$$\theta = 30^\circ$$

Solution

- First, convert the angle to radians

$$\theta = 30 \times \pi / 180$$

$$\theta = 0.52 \text{ rad}$$

- Then convert angular displacement to linear displacement

$$l = r\theta$$

$$l = (0.34 \text{ m})(0.52 \text{ rad})$$

$$l = \mathbf{0.1768 \text{ m}}$$

Part 02

Angular velocity

How fast the angle changes — and what that means at the rim

Again, a scalar and a vector

	ANGULAR SPEED Σ	ANGULAR VELOCITY Ω
What it is	Angular distance covered, divided by the time taken	Angular displacement occurring in a given time
Type	Scalar	Vector
Equation	$\sigma = \Delta\varphi / \Delta t$	$\omega = \Delta\theta / \Delta t$
SI unit	rad/s	rad/s

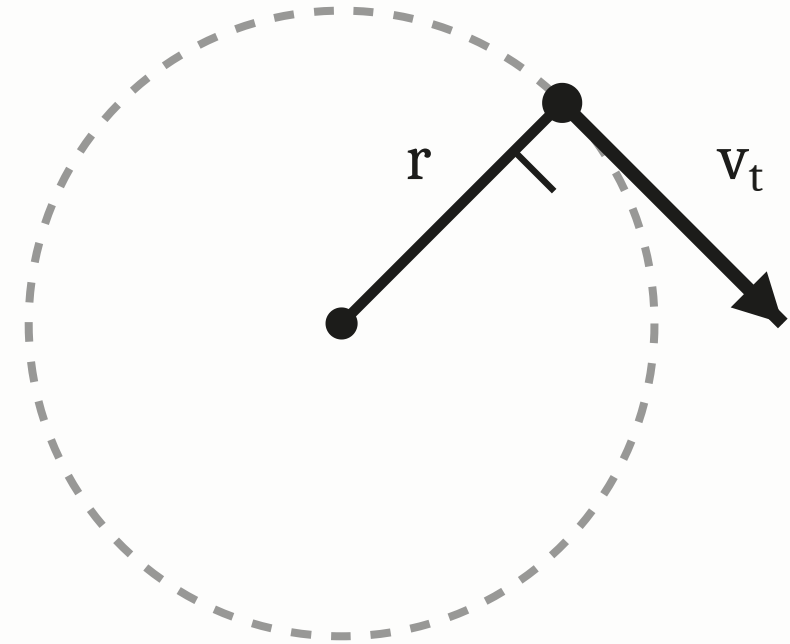
Tangential velocity

The linear velocity of a point travelling an angular path

$$V_t = r\omega$$

v_t in m/s, r in m, ω in rad/s

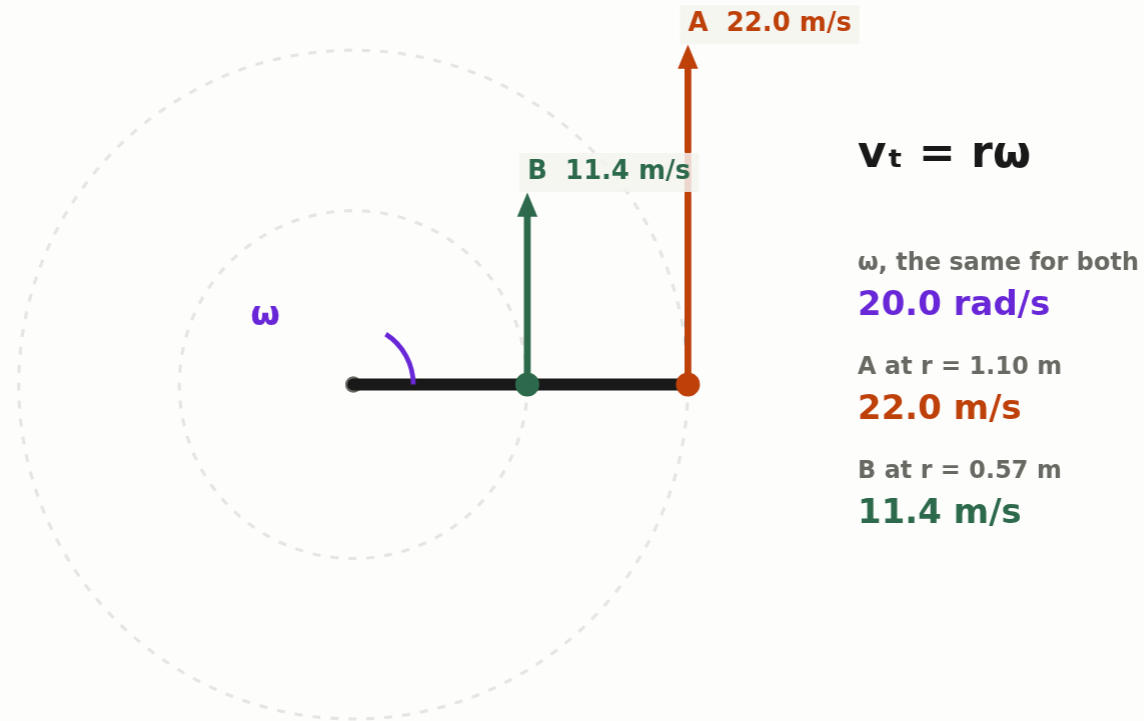
- For a given ω , linear velocities differ with distance from the axis
- The angular velocities of the hands and the bat tip are the **same**; their linear velocities are **not**
- Although it results from angular motion, tangential velocity is a form of **linear** velocity
- Its direction is a straight line **tangent to the circular path**



v_t is always perpendicular to r

Tangential velocity

Same ω everywhere — press play and watch the two arrows



$\omega = 20.0 \text{ rad/s}$ · at $r = 1.10 \text{ m}$ the tip moves at **22.0 m/s**, at $r = 0.57 \text{ m}$ only **11.4 m/s** · the arrow is tangent to the path, so it points where the point would go if it let go right now



Angular velocity
 ω



20.0
rad/s

Example Problem

What is the linear velocity of a point on a bicycle wheel (radius = 34 cm) if the wheel rotates 30° in 0.2 s?

Given:

$$r = 0.34 \text{ m}$$

$$\theta = 30^\circ$$

$$\Delta t = 0.2 \text{ s}$$

Solution

- Convert to radians

$$\theta = 30 \times \pi / 180$$

$$\theta = 0.52 \text{ rad}$$

- Solve for angular velocity

$$\omega = \Delta\theta / \Delta t$$

$$\omega = (0.52 - 0) / 0.2$$

$$\omega = \mathbf{2.6 \text{ rad/s}}$$

- Convert angular to tangential velocity

$$v_t = \omega r$$

$$v_t = (2.6 \text{ rad/s})(0.34 \text{ m})$$

$$v_t = \mathbf{0.884 \text{ m/s}}$$

0.884 m/s is about 3 km/h, which is a very slow wheel — 30° in a fifth of a second is barely moving. The arithmetic is worth sanity-checking against something physical.

Part 03

Angular acceleration

Speeding up, slowing down — and simply turning

Angular acceleration

The rate of change of angular velocity

$$\alpha = \Delta\omega / \Delta t$$

- Represented by the Greek letter **alpha, α**
- The change in angular velocity occurring over a given period of time
- A **vector** quantity
- SI unit: **rad/s²**

$$a_t = \alpha r$$

The component of linear acceleration **tangent** to the circular path is the tangential acceleration

Notice that **r is the common factor** in all three: $l = r\theta$, $v_t = r\omega$, $a_t = r\alpha$. Changing the radius changes every linear quantity in the same proportion.

Example Problem, and solution

What is the tangential acceleration of a point on a bicycle wheel (radius = 34 cm) if the wheel's angular velocity changes from 0 to 2.6 rad/s in 0.2 s?

- First solve for the angular acceleration

$$\alpha = \Delta\omega / \Delta t$$

$$\alpha = (2.6 - 0) / 0.2$$

$$\alpha = \mathbf{13 \text{ rad/s}^2}$$

- Then convert to tangential acceleration

$$a_t = \alpha r$$

$$a_t = (13 \text{ rad/s}^2)(0.34 \text{ m})$$

$$a_t = \mathbf{4.42 \text{ m/s}^2}$$

Exactly the same shape as the velocity problem: divide by time to get the angular quantity, multiply by r to get the linear one.

The bicycle wheel, all the way through

Every step is either $\times r$ or $\div \Delta t$



θ in radians

$$30^\circ \times \pi/180 = \mathbf{0.524 \text{ rad}}$$

arc length $l = r\theta$

$$0.34 \times 0.524 = \mathbf{0.1780 \text{ m}}$$

angular velocity $\omega = \Delta\theta/\Delta t$

$$0.524 / 0.20 = \mathbf{2.62 \text{ rad/s}}$$

tangential velocity $v_t = r\omega$

$$0.34 \times 2.62 = \mathbf{0.890 \text{ m/s}}$$

angular acceleration $\alpha = \Delta\omega/\Delta t$

$$2.62 / 0.20 = \mathbf{13.1 \text{ rad/s}^2}$$

tangential acceleration $a_t = \alpha r$

$$13.1 \times 0.34 = \mathbf{4.45 \text{ m/s}^2}$$

a wheel of $r = \mathbf{0.34 \text{ m}}$ turning $\mathbf{30^\circ}$ in $\mathbf{0.20 \text{ s}}$ · $\omega = \mathbf{2.62 \text{ rad/s}}$ · $v_t = \mathbf{0.890 \text{ m/s}}$ · $a_t = \mathbf{4.45 \text{ m/s}^2}$ · every step is either $\times r$ or $\div \Delta t$ — that is the entire lecture in one column

θ

l

ω

v_t

α

a_t

Angle turned



$\mathbf{30^\circ}$

Centripetal acceleration

Turning is accelerating, even at constant speed

- If something is travelling round a circle then an external force or acceleration is required to change its direction — **Newton's first law**
- That acceleration is **centripetal**, meaning centre-seeking
- The force producing it points **toward the centre**
- It is also called **radial** acceleration

$$a_r = \omega^2 r$$

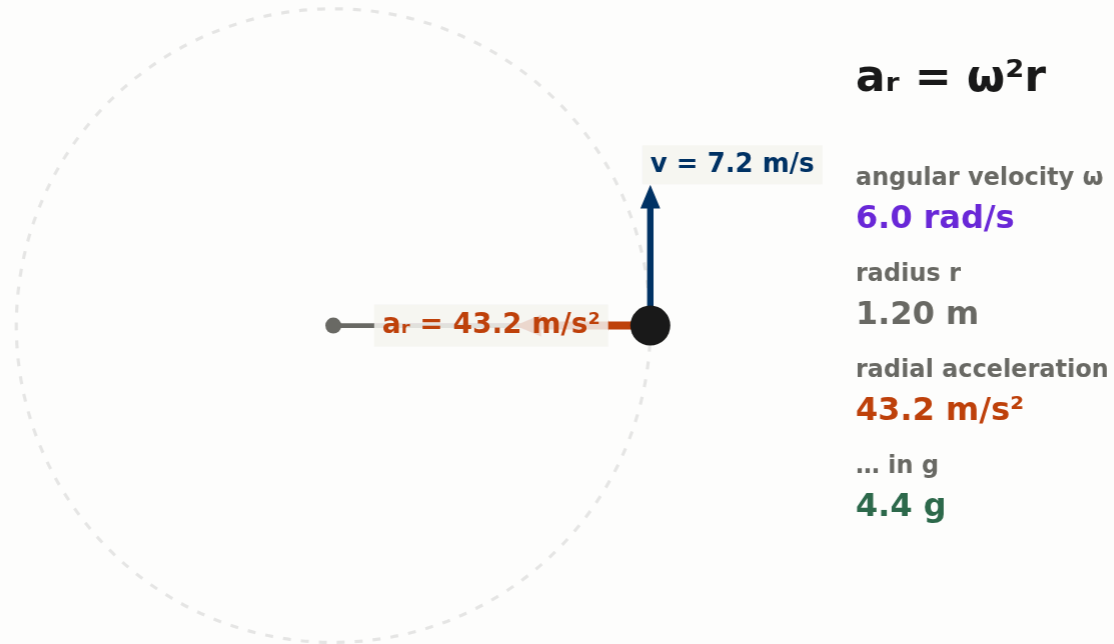
To keep an object at angular velocity ω on a circle of radius r , an inward acceleration of a_r is required



The wire supplies the centre-seeking force

Centripetal acceleration

Press play — then cut the wire



$\omega = 6.0 \text{ rad/s}$, $r = 1.20 \text{ m} \rightarrow a_r = 43.2 \text{ m/s}^2$ (4.4 g) · double ω and the radial acceleration **quadruples**; double r and it only doubles

▶ Play

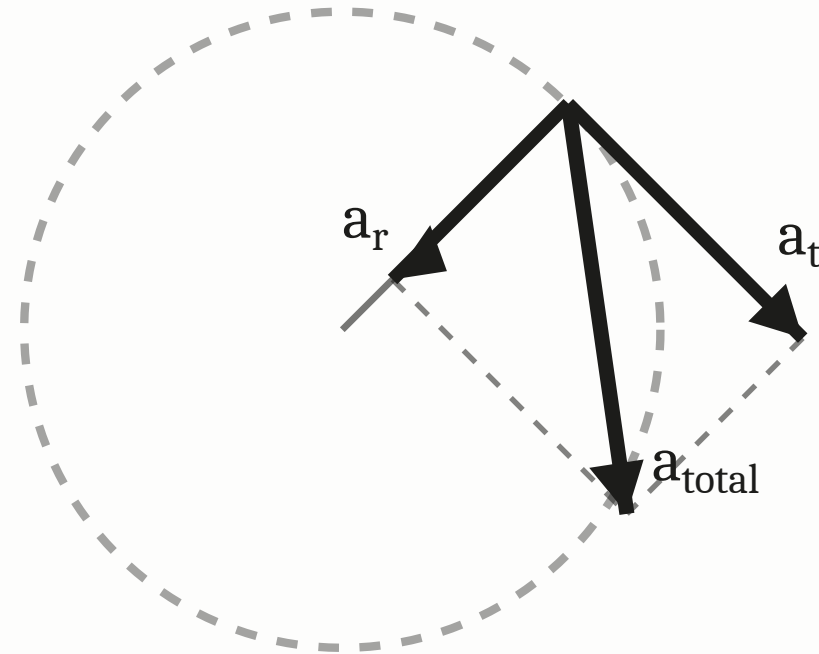
cut the wire

Total linear acceleration

Two perpendicular components, one resultant

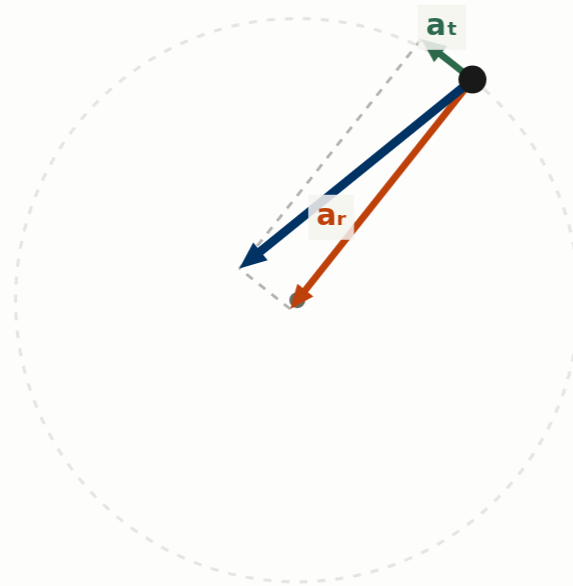
- When an object rotates there are **two** linear accelerations acting:
- $\mathbf{a}_t$, tangential — because it is speeding up or slowing down
- $\mathbf{a}_r$, radial — because it is turning
- Both are **linear** accelerations, even though both arise from angular motion
- They are perpendicular, so they combine by **Pythagoras**

$$a_{\text{total}}^2 = a_r^2 + a_t^2$$



Total linear acceleration

Change ω and α separately, and watch which one matters



$$a_{\text{total}}^2 = a_r^2 + a_t^2$$

tangential $a_t = \alpha r$

9.6 m/s²

radial $a_r = \omega^2 r$

43.2 m/s²

total

44.3 m/s²

the radial part is 98% of the total

$\alpha = 8.0 \text{ rad/s}^2$, $\omega = 6.0 \text{ rad/s}$, $r = 1.20 \text{ m} \rightarrow a_t \text{ 9.6, } a_r \text{ 43.2, total 44.3 m/s}^2$ · once ω is anything much above a few rad/s the radial term swamps the tangential one, because it is squared

Angular velocity

ω



6.0 rad/s

Example Problem



During the hammer throw an athlete changes the angular velocity of the hammer (1.2 m radius) from **0 to 12 rad/s** in **1.5 seconds**.

What is the total acceleration of the hammer?

Given:

$$t = 1.5 \text{ s}$$

$$r = 1.2 \text{ m}$$

$$\omega_i = 0 \text{ rad/s}$$

$$\omega_f = 12 \text{ rad/s}$$

To get the total we need **both** the tangential and the radial accelerations, so this is three calculations and then a Pythagoras.

Solution

- Angular acceleration

$$\alpha = \Delta\omega / t = (12 - 0) / 1.5$$

$$\alpha = \mathbf{8 \text{ rad/s}^2}$$

- Tangential acceleration

$$a_t = \alpha r = (8)(1.2)$$

$$a_t = \mathbf{9.6 \text{ m/s}^2}$$

- Radial acceleration

$$a_r = \omega^2 r = (12)^2(1.2)$$

$$a_r = \mathbf{172.8 \text{ m/s}^2}$$

- Total, by vector composition

$$a_{\text{total}} = \sqrt{(9.6^2 + 172.8^2)}$$

$$a_{\text{total}} \approx \mathbf{173 \text{ m/s}^2}$$

173 m/s² is about **17.6 g**. On a 7.26 kg hammer that is roughly **1260 N** through the wire and the thrower's arms — which is why the event looks the way it does.

Summary of concepts and equations

Three angular quantities, three linear partners, one radius

	ANGULAR	UNIT	LINEAR PARTNER	UNIT
Displacement	$\theta = \text{angle}_2 - \text{angle}_1$	rad	$l = r\theta$	m
Velocity	$\omega = \Delta\theta / \Delta t$	rad/s	$v_t = r\omega$	m/s
Acceleration	$\alpha = \Delta\omega / \Delta t$	rad/s ²	$a_t = r\alpha$	m/s ²
Turning	—		$a_r = \omega^2 r$	m/s ²
Resultant	—		$a_{\text{total}} = \sqrt{a_r^2 + a_t^2}$	m/s ²

Read the table across: every row is "divide by time to go down the angular column, multiply by r to cross to the linear one". The only row without an angular partner is radial acceleration, because turning has no angular equivalent — it is what angular motion *does* to a linear path.

Part 04

Measuring angular motion

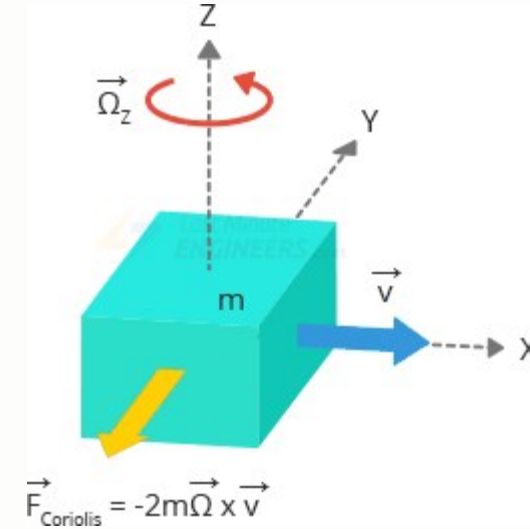
Gyroscopes, the Coriolis effect, and a racing wheelchair

The Coriolis effect

How a sensor with no moving parts detects rotation

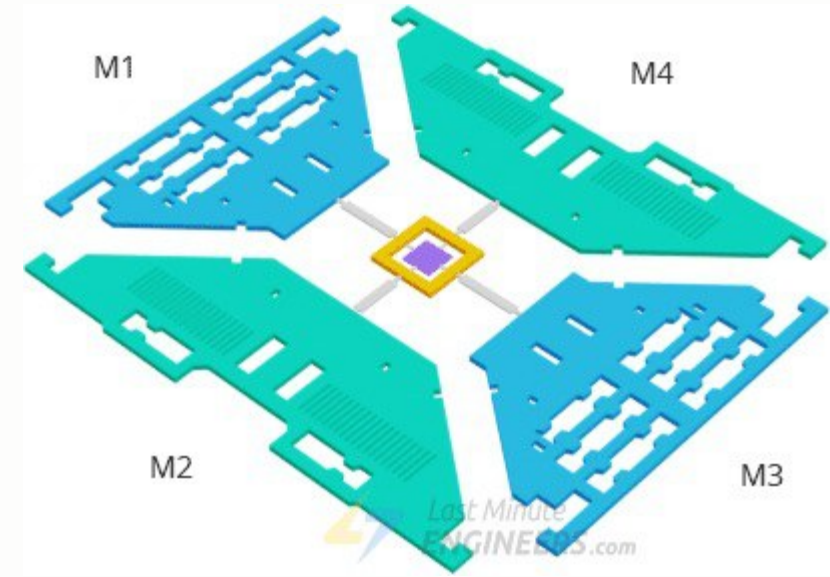
- When a mass **m** moves with velocity **v** and an external angular rate **Ω** is applied...
- ...the Coriolis effect generates a force that pushes the mass **perpendicular** to its motion
- The size of that displacement is **directly related to the angular rate**
- So measuring the sideways push measures the rotation

$$\vec{F}_{\text{Coriolis}} = -2m\vec{\Omega} \times \vec{v}$$



Four masses, kept permanently vibrating

- A **proof mass** in four parts (M1–M4) is held in continuous oscillation, moving inward and outward in the horizontal plane
- Rotate the structure and the Coriolis force **changes that vibration**
- Two masses oscillating in opposite directions get pushed in opposite directions, so they move **apart**
- That changes the **capacitance** between them — the system's ability to store charge — and the change is converted to a voltage

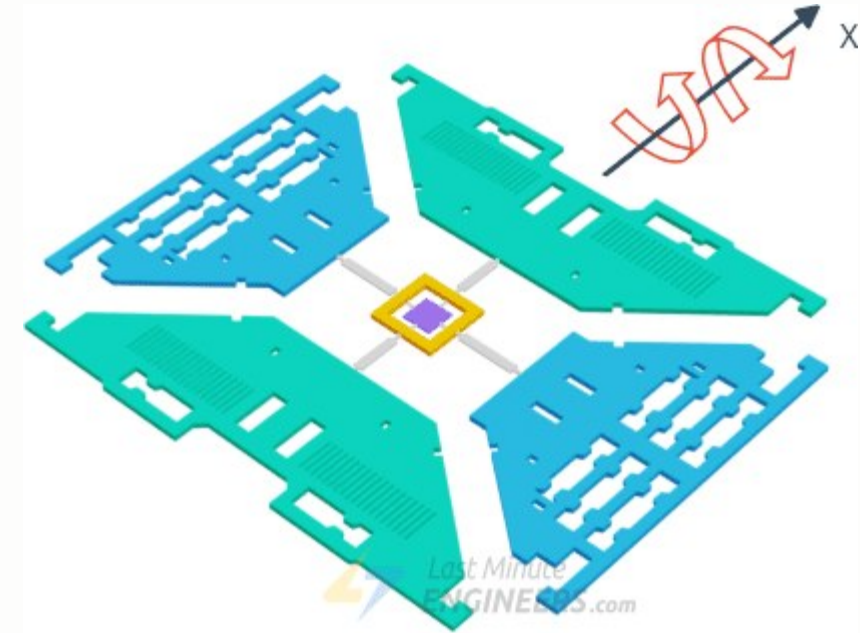


Three axes, three modes

Roll, pitch and yaw come from which masses move

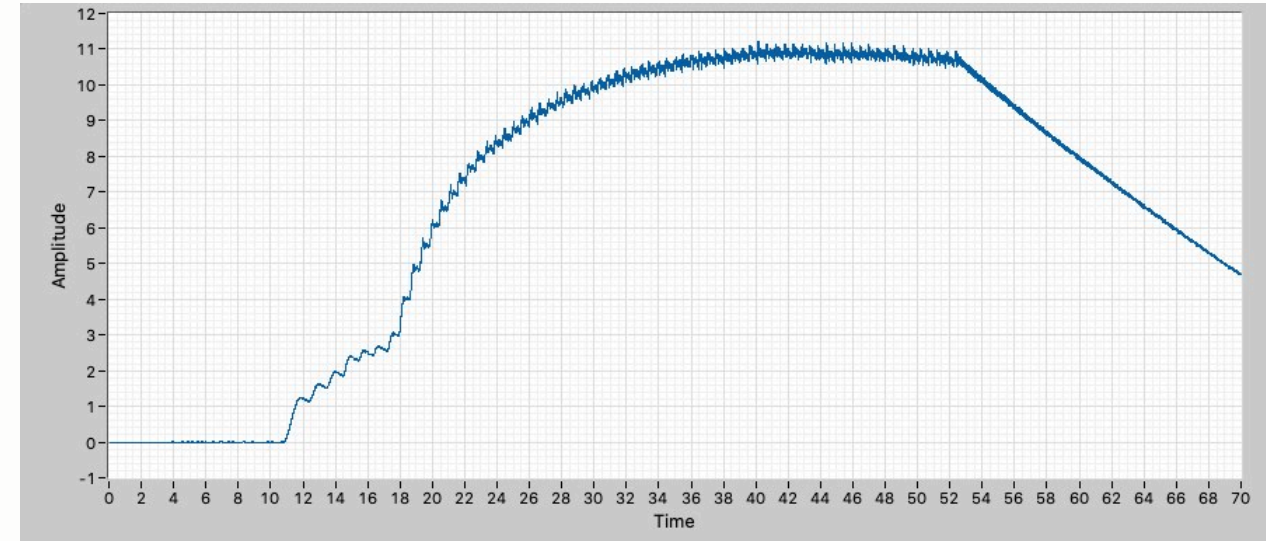
Roll	Angular rate along X . M1 and M3 move up and down out of the plane
Pitch	Angular rate along Y . M2 and M4 move up and down out of the plane
Yaw	Angular rate along Z . M2 and M4 move horizontally in opposite directions

Whenever the Coriolis effect is detected, the constant motion of the driving mass causes a change in capacitance (ΔC) that the sensing structure converts into a voltage signal.



A wheel-mounted sensor

Wheelchair racing metrics from an IMU on the wheel



- The gyroscope reports the wheel's angular velocity, in **degrees per second**
- Every push is visible as a step in the rise
- The trace shows a start, a build, a plateau at top speed, and the slowing at the end
- Nothing here is a speed yet — turning it into one needs the radius

From degrees per second to km/h

Three conversions, and the middle one is this lecture

deg/s → rad/s

$\times \pi / 180$

Because every equation needs radians

rad/s → m/s

$$v_t = r\omega$$

The wheel is 66 cm across, so $r = 0.33$ m

m/s → km/h

$\times 3600 / 1000$

That is just $\times 3.6$

Only the middle step needs anything you did not already know. The sensor never measures speed — it measures **turning**, and the radius is what turns one into the other.

From degrees per second to km/h

Move the sensor reading and watch the chain



$v_t = r\omega$ is the only step that needs the wheel
a 66 cm wheel, so $r = 0.33 \text{ m}$

600 deg/s → 10.47 rad/s → 3.46 m/s → 12.4 km/h · the sensor never measures speed — it measures turning, and the radius is what turns one into the other

first push 200

cruising 600

sprint 1000

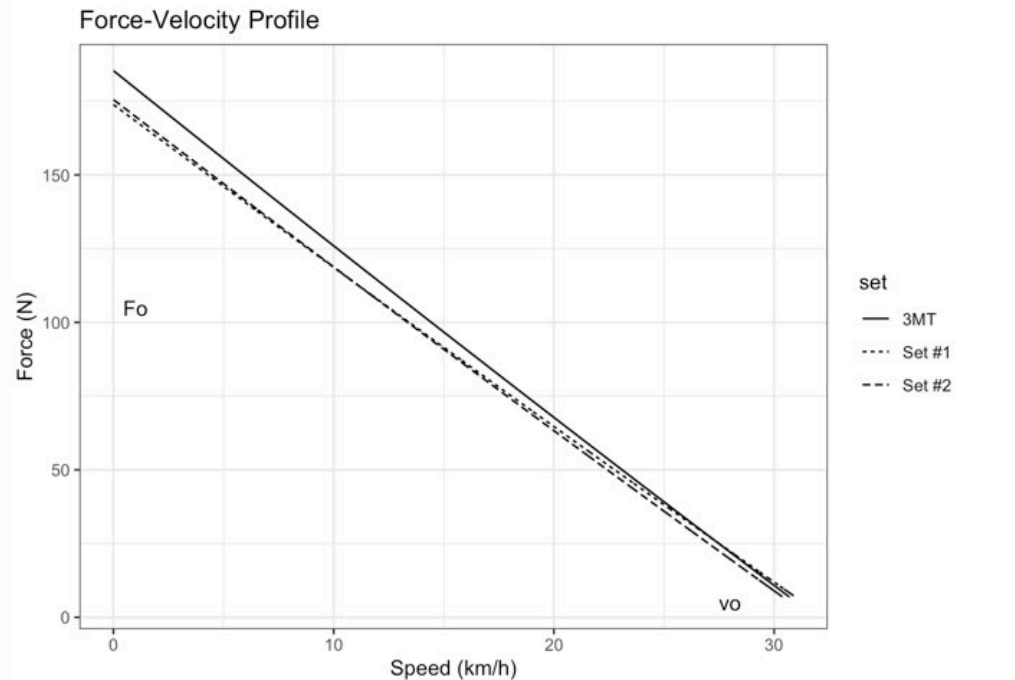
Gyroscope
reading



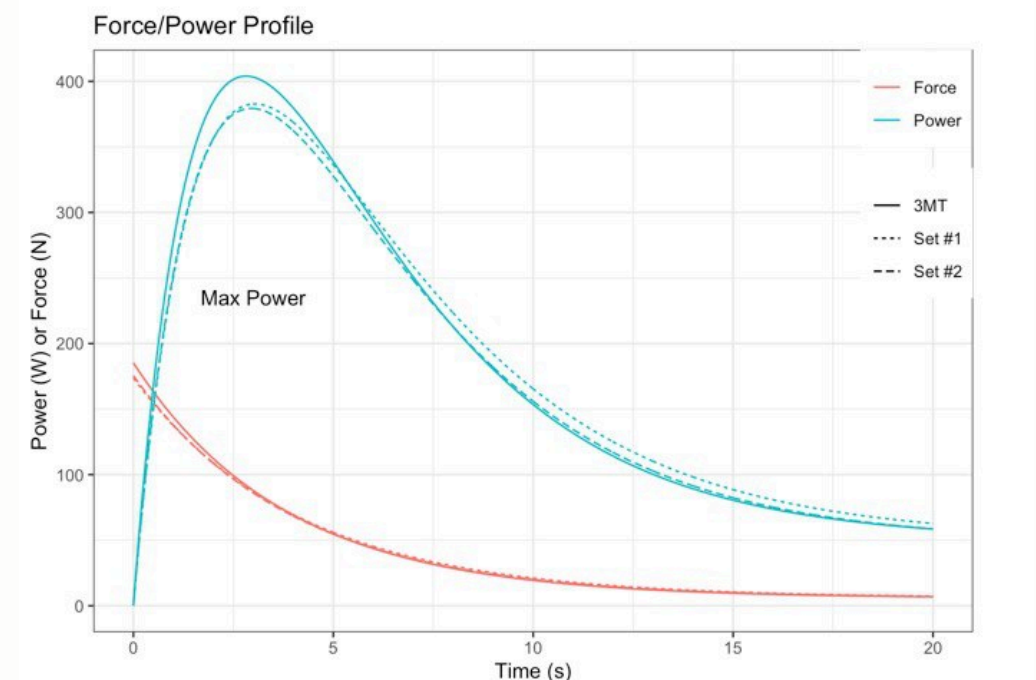
600
deg/s

What you can get from it

Velocity over time is the raw material for everything else



- Differentiate the velocity and you have acceleration, and with mass, force
- Force times velocity gives **power**, peaking at an intermediate speed



- Plot force against velocity across a sprint and you recover a **force-velocity profile** — F_0 and v_0 for the athlete in their chair
- Push-by-push metrics — frequency, contact time, impulse per push — come from the same trace

ANGLES

Absolute angles are measured from a fixed reference, relative angles between two segments. Distance φ is a scalar, displacement θ a vector. Calculations use **radians**, where $\theta = l/r$.

RATES

$\omega = \Delta\theta/\Delta t$ and $\alpha = \Delta\omega/\Delta t$. Every angular quantity has a linear partner through the radius: $l = r\theta$, $v_t = r\omega$, $a_t = r\alpha$.

TURNING

Circular motion needs a centre-seeking acceleration $a_r = \omega^2 r$, even at constant speed. With a_t it gives the total linear acceleration by Pythagoras.

If you remember one thing: **the angle is shared, the distance is not**. Every point on a rotating body turns through the same θ , and the radius decides what that means in metres, metres per second, and metres per second squared.

EPHE 341 · BIOMECHANICS

Questions?

Next up: general kinematics · Further
reading: McGinnis Chpt 6, pg 147–158